

What Does the Weight Norm Control in Grokking? Logit-Scale Mediation under Cross-Entropy

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Abstract

Grokking is generalization that appears long after a network has fit its training data. The grokking delay grows with the weight norm, and holding the norm fixed with a clamp reproduces a clean dose response, which has invited the reading that the weight norm sets the timescale. We show that this reading is incomplete. The clamp holds the scalar norm fixed by rescaling the weights, but rescaling also raises the logit scale, so the norm and the logit scale move together and the clamp does not separate them. We separate them with a non-trainable output temperature that divides the logits before the loss and is not part of the norm. Holding the total weight norm fixed and varying only this temperature slides the delay across the full range produced by raising the norm, and matching the effective logit scale back to its baseline value recovers 0.83 (95% CI [0.82, 0.85]) and 0.89 ([0.88, 0.91]) of the norm-induced delay at two moduli. Across a grid of norms and temperatures the delay collapses onto the effective logit scale, which alone explains 97% of its variance while the norm dose adds 1–2%. Under cross-entropy, then, the weight-norm effect on the delay runs through the logit scale and the softmax saturation it causes, not through the scalar norm itself. The same intervention does nothing of the sort under mean-squared error: there the effective logit scale is pinned near one and cannot be moved, the norm effect is about half the size, and it is carried by a separate route. The mechanism is therefore loss-dependent. A control shows the temperature acts on the delayed-generalization phase and not on memorization time, ruling out a gradient-magnitude artifact, an independent float64 audit reaches the same softmax-saturation channel by precision rather than by intervention, and a no-LayerNorm transformer reproduces the effect. Forking arms from one identical state shows the delay tracks the held norm value, not the rescaling operation, closing the clamp-artifact objection. Claims are confined to the ℓ_2 /weight-decay regime on modular-arithmetic networks.

1 Introduction

When a neural network keeps training after it has fit its training set, it usually stops improving. Grokking is the striking exception: on certain algorithmic tasks, test accuracy stays at chance for many thousands of steps after the training loss has collapsed, then rises sharply to near perfect (Power et al., 2022). The phenomenon is robust and reproducible; the open question is no longer *whether* it happens but *what controls the delay*.

One variable is repeatedly implicated: the weight norm. Grokking coincides with weight-norm decay (Liu et al., 2023), the delay lengthens when the norm is held high, and a recent line treats norm minimization on the zero-loss manifold as the organizing principle (Musat, 2025; Boursier et al., 2025). If the norm is clamped to a chosen value throughout training, the grokking delay

follows a clean exponential dose response in the held norm. It is tempting to read this as the weight norm setting the timescale of grokking.

We argue that this reading conflates two things the clamp changes together. Holding $\|W\|$ fixed requires rescaling the weight matrices, and rescaling also raises the logits, which in cross-entropy pushes the softmax toward saturation. So when a higher held norm lengthens the delay, we cannot yet tell whether the scalar norm is the operative variable or whether it is the logit scale that the rescaling drags along. The distinction matters: the first says grokking is governed by a geometric quantity, the second points at a function-space quantity already implicated by the numerical-stability account of grokking (Prieto et al., 2025).

We separate the two with an intervention. We hold the total weight norm fixed with the clamp and add a non-trainable output temperature τ that divides the logits before the loss. Because τ is not a weight, it is not part of $\|W\|$; at a clamped norm it tunes the effective logit scale while leaving the norm fixed. The result, across two moduli, is unambiguous under cross-entropy: varying τ alone slides the grokking delay across the entire range produced by raising the norm, and the baseline and the norm-raised runs lie on a single curve of delay against effective logit scale. Matching the logit scale back to baseline recovers 0.83 and 0.89 of the norm-induced delay. The weight-norm effect on the delay is, to that extent, the logit-scale effect.

The mechanism is loss-dependent. Under mean-squared error the effective logit scale at grokking is pinned near one because the regression target fixes it, τ cannot move it, and the norm effect, while still present, is about half the size and is not carried by the logit scale. A control rules out the obvious artifact: τ leaves the memorization time essentially unchanged and acts almost entirely on the delayed-generalization phase, so the effect is about grokking specifically and not about training speed.

Our contributions:

- A temperature-mediation test that separates the scalar weight norm from the effective logit scale at a fixed clamped norm, and the finding that under cross-entropy the norm-induced grokking delay is recovered primarily (~ 0.85 , tight bootstrap CI, two moduli) by restoring the logit scale.
- A data collapse: across a grid of norms and temperatures, the delay is a function of the effective logit scale alone ($R^2 = 0.97$), with the norm dose adding 1–2% beyond it.
- A clean loss dissociation: the logit-scale channel is active under cross-entropy and absent under mean-squared error, so the weight-norm dependence of grokking is not a single mechanism.
- A memorization control showing the temperature acts on the delay and not on memorization, a float64 audit that reaches the same softmax-saturation channel independently, and qualitative corroboration in a no-LayerNorm transformer.
- A same-state test that forks arms from one identical state and shows the delay tracks the held norm value, not the clamp’s rescaling operation, closing the rescaling-artifact objection for this setting.
- Honest scope: the result localizes the proximal variable under cross-entropy; it does not characterize the mean-squared-error route, and the quantitative estimates are for MLPs at two moduli.

2 Setup

Task and model. We study modular addition, inputs $(a, b) \in \{0, \dots, p-1\}^2$ and target $(a + b) \bmod p$, a p -way classification, for $p \in \{43, 59, 67, 97, 113\}$. The dataset is all p^2 pairs; a fixed fraction $\alpha = 0.40$ is the training split, drawn per seed by a seeded permutation. The model is a two-layer MLP: each of a, b is embedded through a shared $E \in \mathbb{R}^{p \times d}$ ($d = 128$), the two embeddings are concatenated and passed through a linear layer $W_1 \in \mathbb{R}^{2d \times H}$ ($H = 256$) with a GeLU nonlinearity, then a linear readout $W_2 \in \mathbb{R}^{H \times p}$. Optimizer is AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$, learning rate 10^{-3} , weight decay $\lambda = 1.0$ unless varied), full-batch unless noted, 12 seeds per cell. The weight norm is $\|W\| = \sqrt{\|E\|_F^2 + \|W_1\|_F^2 + \|W_2\|_F^2}$, biases excluded.

The clamp. After each optimizer step from $t \geq t_{\text{int}}$, the weight matrices are rescaled by a single scalar so that $\|W\| = \rho w_c$ exactly, where w_c is the norm at grokking measured in a free control run and ρ is the dose. The clamp is engaged before memorization in every cell used here, so the intervention is active throughout the relevant dynamics.

Effective logit scale. The central variable of this paper needs a precise definition. For a configuration with temperature τ , the effective logits are the post- τ logits the loss actually sees. The effective logit scale is the mean, over all p^2 input pairs and over seeds, of the L_2 norm of the per-example effective logit vector, read at the grokking step (the logged step nearest the median T_{grok} of that cell). In the mediation analysis it is normalized by the baseline ($\rho = 1$, $\tau = 1$) value so the axis is dimensionless and comparable across moduli.

Temperatures and metrics. The temperature τ divides the logits before the softmax (CE) or the squared error (MSE); $\tau = 1$ recovers the standard model and τ is held fixed within a run. We log test accuracy (grokking time T_{grok} is the first step with test accuracy ≥ 0.90 , median over seeds), training accuracy (memorization time T_{mem} , ≥ 0.99), the per-group weight norms, the effective logit scale, and the softmax-collapse rate of [Prieto et al. \(2025\)](#).

3 The fixed-norm exponential law (the dose response)

Holding the norm fixed and sweeping the dose ρ yields a clean dose response between the held norm and the grokking time. Across all five moduli the relationship is exponential, $T_{\text{grok}} \propto \exp(\alpha \|W\|)$, with a per-modulus slope α . We compare this against a power law and against a saddle-node form $T \propto (w_0 - \|W\|)^{-1/2}$ (the signature a fold or critical-slowness bifurcation would produce) by AIC. The exponential is generally selected (Table 1): decisively at four of the five moduli (ΔAIC 4.6–6.4), and weakly at $p = 43$ (ΔAIC 0.4, essentially tied with the power-law), which we report rather than smooth over. We read this as a robust empirical regularity of the fixed-norm dose response, not a law derived from first principles, and we use it here only as the readout against which the interventions below are measured. The slopes are fit on the softmax-collapse-free range ($\rho \leq 1.15$; §5).

This dose response is the phenomenon to be explained. It establishes that raising the held norm lengthens the delay; it does not, on its own, say what the norm acts through. That is the question of the next section.

Table 1: Fixed-norm exponential law $\ln T_{\text{grok}} = c + \alpha \|W\|$ per modulus, fit on the softmax-collapse-free range ($\rho \leq 1.15$), majority-grok cells only. ΔAIC is the exponential’s margin over the better of the power-law and saddle-node forms (positive favors the exponential).

p	n	α	R^2	ΔAIC	SC-free norm range
43	6	0.173	0.987	0.4 (weak)	45–58
59	7	0.140	0.991	6.4	46–62
67	7	0.126	0.982	5.5	48–65
97	7	0.094	0.957	4.6	56–75
113	7	0.087	0.959	4.6	60–81

4 What the weight norm acts through

4.1 A temperature-mediation test

The clamp holds $\|W\|$ fixed by rescaling the weights, but that rescaling raises the logits, and in cross-entropy larger logits saturate the softmax. So the dose response of §3 does not separate the scalar norm from the logit scale it sets. We separate them with a non-trainable output temperature τ that divides the logits before the loss and is not counted in $\|W\|$. At a clamped norm, τ tunes the effective logit scale while leaving the norm fixed.

The design is a three-condition mediation. Baseline ($\rho = 1.0$, $\tau = 1$) gives delay T_0 ; raising the norm ($\rho = 1.15$, $\tau = 1$) gives T_1 ; and at $\rho = 1.15$ we sweep τ upward, which lowers the effective logit scale at fixed norm. If the norm effect runs through the logit scale, setting τ so that the effective logit scale matches the baseline should recover the baseline delay.

Under cross-entropy this is what happens (Fig. 1a, Table 2). The norm-up delay is $3.4\times$ the baseline at both moduli. Sweeping τ from 1.0 to 1.7 slides the delay monotonically back down, from $3.4\times$ to below baseline, and the cells trace a single curve of delay against effective logit scale on which the baseline and norm-up points both lie. The effective logit scale itself moves by about $1.6\times$ across the sweep. Reading off the delay where the effective logit scale equals its baseline value gives a *logit-scale recovery fraction* $(T_1 - T_2)/(T_1 - T_0)$ of 0.83 at $p = 59$ (95% bootstrap CI [0.82, 0.85]) and 0.89 at $p = 97$ ([0.88, 0.91]); the CI is a paired resample over the twelve seeds and reflects seed variability in T_{grok} , with the logit curve held fixed. We call it a recovery fraction rather than a mediation fraction deliberately: it is the share of the norm-up delay that the temperature intervention recovers by restoring the effective logit scale, an intervention-level quantity, not a regression-based natural-indirect-effect estimate. To that extent the weight-norm effect on the delay is the logit-scale effect: the norm matters because it sets the logit scale, which sets the softmax saturation. This is the same channel the audit of §5 probes, reached here by intervention rather than by precision. The next subsection shows the relationship is not specific to this one norm level: across a grid of norms and temperatures, the delay collapses onto the effective logit scale.

4.2 A data collapse across the norm–temperature grid

The single-norm recovery fraction asks what happens at one norm level. A stronger question is whether the delay is a function of the effective logit scale *regardless* of how that scale was reached, whether by raising the norm or by lowering the temperature. We test this on a grid of four norm doses ($\rho \in \{1.00, 1.05, 1.10, 1.15\}$) crossed with three temperatures ($\tau \in \{1.0, 1.3, 1.7\}$), twelve cells per modulus, all grokking 12/12. Figure 2 plots T_{grok} against the effective logit scale at grokking, colored by ρ : the cells fall on a single log-linear curve, and cells of different ρ at the same effective logit scale have the same delay.

Table 2: Temperature mediation at $\rho = 1.15$ (medians over 12 seeds). The logit-scale recovery fraction is the share of the norm-up delay recovered by matching the effective logit scale back to baseline; it is reported only for cross-entropy, where the logit scale is a usable knob.

loss	p	T_0	T_1	norm-up	recovery fraction
CE	59	3675	12600	$3.43\times$	0.83 [0.82, 0.85]
CE	97	1775	6050	$3.41\times$	0.89 [0.88, 0.91]
MSE	59	1175	2225	$1.89\times$	n/a (channel inactive)
MSE	97	850	1412	$1.66\times$	n/a (channel inactive)

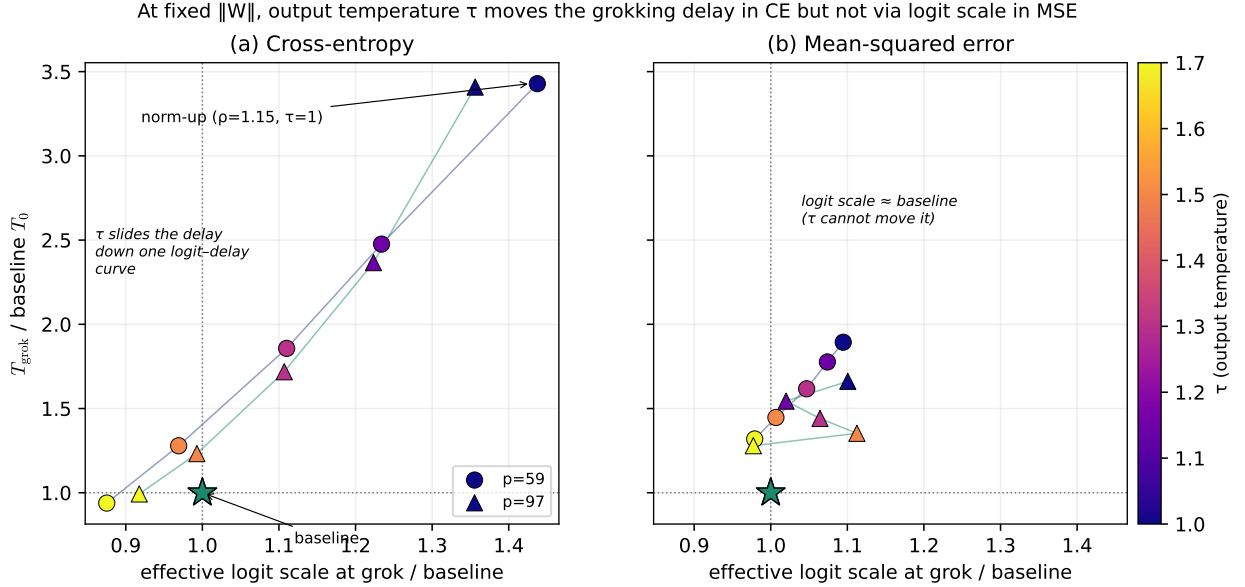


Figure 1: At a fixed clamped norm, varying only the output temperature τ . **(a)** Under cross-entropy the cells trace one curve of delay against effective logit scale: the baseline (star) and the norm-up point ($\tau = 1$, top right) lie on it, and increasing τ slides the delay back down toward baseline. About 83–89% of the norm-up delay is recovered by matching the logit scale (two moduli). **(b)** Under mean-squared error the effective logit scale at grokking is pinned near baseline and τ cannot move it; the residual norm effect (vertical spread) is not along the logit-scale axis. Both moduli overlaid; axes normalized by the baseline cell.

We quantify the collapse by regression. A least-squares fit of $\ln T_{\text{grok}}$ on the effective logit scale explains $R^2 = 0.97$ of the variance at both moduli; adding ρ as a second predictor raises R^2 by only 0.02 ($p = 59$) and 0.01 ($p = 97$), so once the effective logit scale is known the norm dose carries almost no additional information about the delay (Table 3). The delay is, to this precision, a function of the effective logit scale alone. We are careful about the causal reading: the x -axis is the *realized* logit scale at grokking and is therefore endogenous, so the collapse is strong organizing evidence, not an intervention on its own; the causal weight rests on the temperature manipulation of §4.1, which sets the logit scale directly at fixed norm. The two are complementary: the intervention shows the channel is causal, and the collapse shows it accounts for essentially all of the norm dependence.

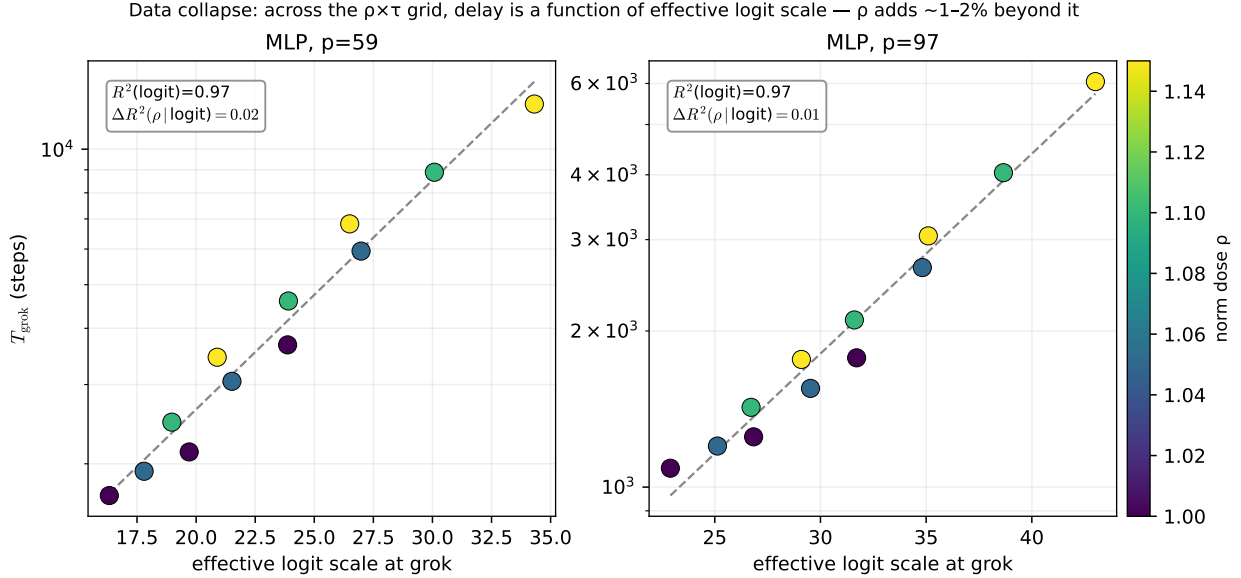


Figure 2: Data collapse across the $\rho \times \tau$ grid (12 cells per modulus, all grok 12/12). T_{grok} against the effective logit scale at grokking, colored by the norm dose ρ . Cells of different ρ but matched effective logit scale share the same delay, so the delay collapses onto the logit scale; the norm dose adds 1–2% of explained variance beyond it.

Table 3: Collapse regression across the $\rho \times \tau$ grid. R^2 of $\ln T_{\text{grok}}$ against the effective logit scale alone, against ρ alone, and against both; $\Delta R^2(\rho | \text{logit})$ is the variance ρ adds beyond the logit scale (near zero indicates collapse).

p	$R^2(\text{logit})$	$R^2(\rho)$	$R^2(\text{both})$	$\Delta R^2(\rho \text{logit})$	verdict
59	0.968	0.407	0.988	0.020	collapse
97	0.973	0.415	0.984	0.011	collapse

4.3 The mechanism is loss-dependent

Mean-squared error dissociates cleanly (Fig. 1b). There the effective logit scale at grokking sits near one and barely moves with τ , because the regression target fixes the output scale, so the temperature cannot act on a logit-scale channel: across $\tau \in [1.0, 1.7]$ the effective logit scale moves about $1.1\times$, against $1.6\times$ under cross-entropy. The norm-up effect under MSE is real but about half the size ($1.7\text{--}1.9\times$ versus $3.4\times$) and is carried by a route that does not pass through the logit scale. So the weight-norm dependence of the delay is not one mechanism. Under cross-entropy it is logit-scale saturation; under mean-squared error it is something else.

We are deliberately conservative about the MSE residual. The temperature still shortens the MSE delay somewhat, but a temperature that divides the logits also changes the scale of the squared-error target, so we do not read this residual as a clean second channel. We report MSE only to establish the dissociation that the logit-scale channel is absent there; we do not assign the residual norm effect to a specific mechanism. This connects to an asymmetry noted by [Prieto et al. \(2025\)](#), who observed that grokking under softmax cross-entropy behaves differently from the MSE setting of [Liu et al. \(2023\)](#), with the softmax-collapse rate approaching one under cross-entropy; our intervention gives that asymmetry a mechanism on the cross-entropy side.

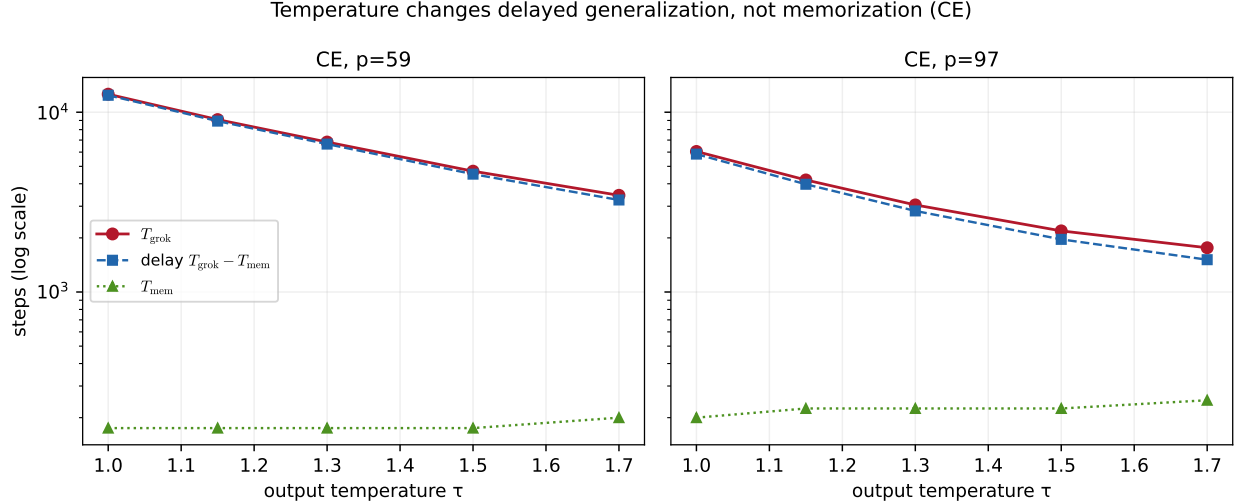


Figure 3: Under cross-entropy, the temperature changes the delayed-generalization phase, not memorization. T_{mem} (bottom) is flat across τ while T_{grok} and the delay $T_{\text{grok}} - T_{\text{mem}}$ fall together. y -axis is log scale.

4.4 The effect is on the delay, not on memorization

A temperature that divides the logits changes both the softmax saturation and the gradient magnitude, so a skeptic can ask whether τ simply speeds up training in general. It does not, under cross-entropy. Splitting the per-seed times (Fig. 3), the memorization time T_{mem} is essentially flat across τ (175 $\rightarrow$ 200 steps at $p = 59$, 200 $\rightarrow$ 250 at $p = 97$), while the delay $T_{\text{grok}} - T_{\text{mem}}$ carries the entire τ effect (100% and 101% of the change in T_{grok}). The temperature acts on the delayed-generalization phase, not on how fast the network fits the training set, so the gradient-magnitude reading does not explain it. (Under MSE the split is less clean: at $p = 97$ about two thirds of the τ effect falls on T_{mem} , a further reason we do not interpret the MSE residual.)

4.5 Corroboration across allocation and architecture

A second intervention corroborates that the scalar norm is not the sole driver. Holding the total norm fixed and shifting mass toward the readout layer (raising $\|W_2\|$ at the expense of $\|E\|$) changes the delay by up to $2.2\times$ at $p = 59$ and $1.7\times$ at $p = 97$. We treat this as supporting rather than primary, because reallocating mass also changes embedding capacity, so it is not a clean logit-scale move; the temperature test is the clean manipulation.

The mechanism also reproduces in a different architecture. We ran the temperature test on a no-LayerNorm transformer at $p = 59$: raising the held norm lengthens the delay, and at the raised norm, increasing the temperature shortens the delay monotonically by $4.7\times$ (16,100 $\rightarrow$ 3,500 steps), confirming that a variable outside the norm controls the delay under cross-entropy in a transformer as it does in the MLP. We report this as qualitative corroboration and deliberately do not quote a recovery fraction for the transformer: its logits grow far larger and more erratically than the MLP’s. The highest-stress cell shows the uncontrolled logit growth of Prieto et al. (2025), with the logit scale climbing past 1000 after grokking, so the effective logit scale at grokking is a noisier proxy there, and matching it back to baseline would require extrapolation. The direction and size of the effect transfer; the precise mediation estimate is MLP-specific. That the transformer’s logits are harder to hold down is itself consistent with the saturation account.

We are also explicit about what the temperature test does not do on its own. It shows that, at

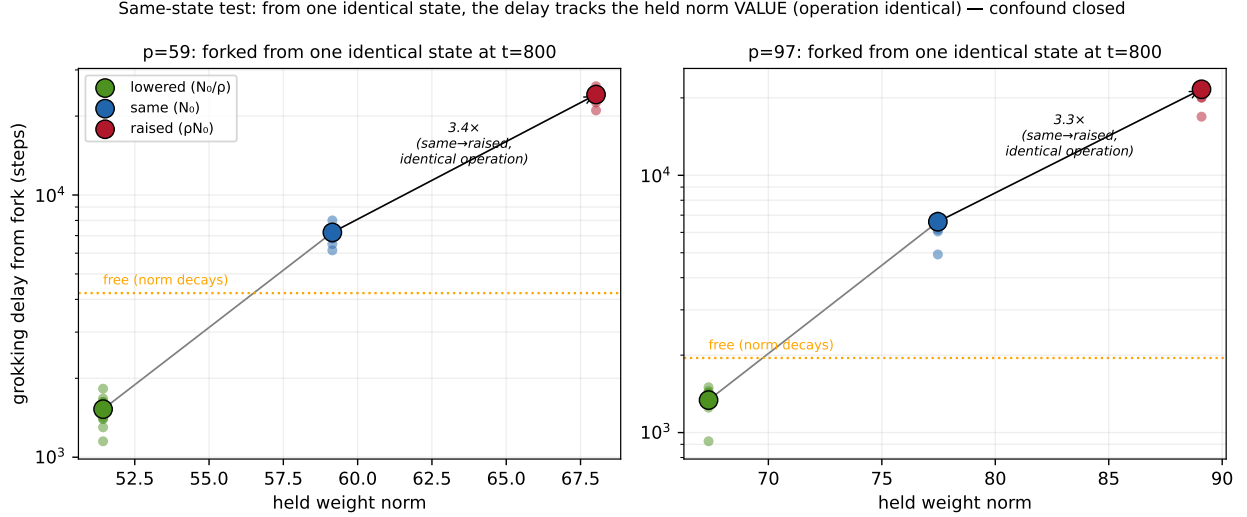


Figure 4: Four arms forked from one identical post-memorization state ($t = 800$). The grokking delay (from the fork) rises monotonically with the held norm value across all twelve seeds. The labelled contrast, clamp-at- N_0 versus clamp-at- ρN_0 , applies the identical clamp operation and differs only in the held value ($3.3\text{--}3.4\times$), so the delay tracks the value, not the rescaling operation. Free (dotted) groks faster because its norm decays after the fork.

a fixed clamped norm, a variable outside the norm controls the delay, which localizes the proximal variable under cross-entropy to the logit scale, and we claim the logit scale is the proximal mediator of the norm effect. The remaining question is whether the clamp’s rescaling operation, rather than the held value, drives the delay; the next subsection settles it.

4.6 A same-state test of the clamp

The clamp sets the norm by rescaling the weights, so one objection survives the temperature test: perhaps the rescaling operation, applied every step, distorts the trajectory, and the delay reflects the operation rather than the held value. We separate the two by forking four arms from one identical state. We train freely to a fork step after memorization and before grokking, then continue from the saved state under four conditions at $\rho = 1.15$: free (no clamp), clamp at the forked norm N_0 (the operation is active but the value is unchanged), clamp at ρN_0 , and clamp at N_0/ρ .

From the identical state, the delay rises monotonically with the held norm value, in all twelve seeds at both moduli (Fig. 4). The decisive contrast is clamp-at- N_0 against clamp-at- ρN_0 . Both apply the same clamp operation from the same state and differ only in the held value, and the raised arm takes $3.36\times$ longer at $p = 59$ (95% CI $[3.27, 3.43]$) and $3.27\times$ at $p = 97$ ($[3.21, 3.38]$). Since the operation is identical in the two arms, the difference is not an artifact of the operation: the delay follows the held value. The ordering across lower, hold, and raise also rules out a one-time rescaling jump, which would perturb the up and down arms the same way instead of ordering them by value. The free arm groks faster than clamp-at- N_0 because its norm decays after the fork, which is the norm decay that the generalizing-shell account associates with grokking (Liu et al., 2023). This closes the rescaling-artifact objection for the cross-entropy MLP setting. Training full-batch leaves the dose response intact, so the effect is not a sampling-noise phenomenon either.

5 A softmax-collapse audit

Because the cross-entropy clamp operates at large logits, the regime in which softmax collapse can occur (Prieto et al., 2025), we audit the law for floating-point artifacts. We re-run the two highest-dose cross-entropy cells at $p = 59$ in float64, holding the norm at the identical absolute values used in float32, and record the softmax-collapse rate (Table 4). At $\rho = 1.15$ the collapse rate is zero in both precisions and the grokking times agree, so the law is precision-robust there. At $\rho = 1.25$ the float32 run shows a 31% collapse rate and groks at 25k steps, whereas the float64 run does not grok within 60k, so the float32 collapse accelerates a spurious transition at extreme norm. We therefore fit the law on the collapse-free range and exclude the affected cell. This audit is convergent with §4.1: softmax collapse is the extreme of the same logit-saturation channel that the temperature test isolates, reached by precision without any intervention.

Table 4: Softmax-collapse audit at $p = 59$: the two highest-dose cross-entropy cells re-run in float64 at held norms identical to float32. T = median grok step; sc = softmax-collapse rate.

ρ	held $\ W\ $	T_{32}	T_{64}	sc ₃₂	sc ₆₄	verdict
1.15	62	12.75k	12.16k	0.00	0.00	precision-robust
1.25	68	25.4k	not reached by 60k	0.31	0.00	float32 SC-confounded

6 Related work

Weight norm and norm minimization. The weight norm has long been central to grokking: Omnigrok ties grokking to the norm decaying into a generalizing shell (Liu et al., 2023), and a recent line characterizes the late dynamics as norm minimization on the zero-loss manifold (Musat, 2025; Boursier et al., 2025), including beyond the Euclidean norm (Tikeng Notsawo et al., 2025). These accounts concern the *endpoint* (which configuration the optimizer selects once training loss is zero). Our question is about the *timescale* and the proximal variable that sets it; we find that under cross-entropy the norm acts on the timescale through the logit scale, so the two are complementary rather than competing.

Logit growth, softmax saturation, and optimization starvation. The account closest to our result is the numerical-stability view of Prieto et al. (2025): cross-entropy training without regularization drives uncontrolled logit growth (“naive loss minimization”), whose extreme is softmax collapse, where floating-point errors zero the gradient and stall learning. Our temperature test isolates the upstream of that channel, the logit scale, and shows it mediates most of the norm-induced delay under cross-entropy, with the collapse of Prieto et al. (2025) as its endpoint (§5). Independently, Beck et al. (2026) show that regularizing logits directly biases linear classifiers toward clustered logits and can induce grokking, further evidence that logit space, not weight norm alone, is the operative arena under cross-entropy. The intervention itself, dividing logits by a fixed temperature, is the temperature scaling of Guo et al. (2017); we use it not for calibration but as a handle on the effective logit scale at fixed norm.

Phase transitions and scaling. A parallel line characterizes grokking as a phase transition along model or data axes: a first-order transition in two-layer networks (Rubin et al., 2024), an information-theoretic transition (Clauw et al., 2024), Ising-style and local-rule descriptions (Hutchinson and Yevick, 2025; Žunkovič and Ilievski, 2024), and finite-size or dimensional scaling (Bi et al., 2026; Wang, 2026b,a). These concern the nature of the transition and its scaling with model or

data size, a different axis from our intervention-level question of which variable gates the delay at fixed data and model.

Other proposed drivers. Grokking has also been attributed to noise-assisted escape and slingshot dynamics (Thilak et al., 2022; Lopatin et al., 2025), to circuit-efficiency reallocation (Nanda et al., 2023; Varma et al., 2023), and to a lazy-to-rich transition (Kumar et al., 2024; Lyu and Li, 2020). Our full-batch result places the delay outside the noise-activated family for this setting, and our structure measure is the Fourier progress measure of Nanda et al. (2023).

Most of this work asks what *produces* grokking and answers with a mechanism that suffices. Our question is narrower: with the weight norm held at a chosen value, which mechanism still moves the delay? The temperature handle lets us put one candidate, the logit scale, in and take it out directly, so the contribution here is not another mechanism but a way to decide, under cross-entropy, what the norm acts through.

7 Limitations

The claims are confined to the ℓ_2 /weight-decay regime on modular-arithmetic networks. The quantitative results (the recovery fraction and the data collapse) are established on MLPs at two moduli; a no-LayerNorm transformer reproduces the effect in direction and size but with noisier logit dynamics, so we do not extend the precise recovery estimate to it, and we have not tested larger-scale settings or other task families (sparse parity, group tasks beyond modular arithmetic). The same-state test (§4.6) addresses the clamp-rescaling confound for the cross-entropy MLP setting, but the matched contrast is at one fork point and one ρ ; we have not mapped the full fork-time dependence. The data collapse uses the realized logit scale at grokking, which is endogenous, so it is organizing evidence and the causal weight rests on the temperature intervention. The mean-squared-error route is identified only negatively: it does not pass through the logit scale, and we do not characterize it further; the residual temperature effect under MSE is partly a memorization-speed effect and is not interpreted. The recovery-fraction CI reflects seed variability in T_{grok} with the logit curve held fixed, and does not include interpolation uncertainty on the logit axis. Two directions follow directly: characterizing the mean-squared-error route, which is left here as not-the-logit-scale, and testing whether the logit-scale mediation holds at transformer scale, where the logit dynamics we saw are already noisier.

8 Conclusion

The weight-norm dependence of the grokking delay is not, under cross-entropy, an effect of the scalar norm. Holding the norm fixed and varying only an output temperature slides the delay across the full norm-up range, and matching the effective logit scale back to baseline recovers about 85% of the delay at two moduli, with the effect falling on the delayed-generalization phase and not on memorization. Across a grid of norms and temperatures the delay collapses onto the effective logit scale, which alone explains 97% of its variance. The weight norm is an upstream handle; the proximal variable is the logit scale, and the channel is the softmax saturation that the numerical-stability account of grokking also points to. The mechanism is loss-dependent: mean-squared error pins the logit scale and the norm effect runs elsewhere. Forking arms from one identical state confirms that the delay follows the held norm value and not the rescaling operation, so the dose response is not a clamp artifact. Beyond grokking, the methodological point is that a causal claim about a scalar quantity such as the weight norm has to be separated from the function-space variables

that rescaling the weights changes along with it; here, once that separation is made, much of what looked like a weight-norm law is a logit-scale law.

Appendix

A Model, task, and exact definitions

Task. Modular addition: inputs $(a, b) \in \{0, \dots, p-1\}^2$, target $(a+b) \bmod p$, a p -way classification, for $p \in \{43, 59, 67, 97, 113\}$ (the dose-response law) and $p \in \{59, 97\}$ (the interventions). The dataset is all p^2 pairs; a fixed fraction $\alpha = 0.40$ is the training split, drawn per seed by a seeded permutation, the remainder held out for test.

Model. A two-layer MLP. Each of a, b is mapped through a shared embedding $E \in \mathbb{R}^{p \times d}$ ($d = 128$); the two embeddings are concatenated and passed through a linear layer $W_1 \in \mathbb{R}^{2d \times H}$ ($H = 256$) with bias b_1 and a GeLU nonlinearity, then a linear readout $W_2 \in \mathbb{R}^{H \times p}$ with bias b_2 . Weight matrices are initialized $\mathcal{N}(0, 1)/\sqrt{\text{fan-in}}$; biases initialized to zero.

Optimizer. AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$), learning rate 10^{-3} , weight decay $\lambda = 1.0$ unless varied. Full-batch gradients unless a minibatch noise axis is specified. Weight decay and the clamp act on E, W_1, W_2 ; biases are unregularized and excluded from $\|W\|$. Twelve seeds per cell.

Weight norm. $\|W\| = \sqrt{\|E\|_F^2 + \|W_1\|_F^2 + \|W_2\|_F^2}$. This is the quantity the clamp holds and the x -axis of the dose response.

The clamp. After each optimizer step with $t \geq t_{\text{int}}$ ($t_{\text{int}} = 500$ for the cross-entropy runs, 50 for the mean-squared-error runs, chosen below memorization so the intervention is active throughout the relevant dynamics), the weight matrices are rescaled by a single scalar so that $\|W\| = \rho w_c$ exactly, where w_c is the median norm at grokking in a free control run and ρ is the dose. The control calibrates $w_c = 54.49$ ($p = 59$) and 65.85 ($p = 97$) under cross-entropy.

Temperature. A non-trainable scalar τ divides the logits before the loss; $\tau = 1$ recovers the standard model. τ is not a parameter and is not part of $\|W\|$, so at a clamped norm it changes the effective logit scale while leaving $\|W\|$ fixed.

Effective logit scale. For a configuration with temperature τ , the effective logits are the post- τ logits the loss sees. The effective logit scale is $\frac{1}{S} \sum_s \frac{1}{p^2} \sum_{(a,b)} \|z_{s,(a,b)}/\tau\|_2$, the mean over all p^2 input pairs and over the $S = 12$ seeds of the L_2 norm of the per-example effective logit vector z/τ , read at the grokking step (the logged step nearest the median T_{grok} of the cell). In the mediation analysis it is normalized by the baseline ($\rho = 1$, $\tau = 1$) value.

Times. T_{mem} is the first step with train accuracy ≥ 0.99 ; T_{grok} is the first step with test accuracy ≥ 0.90 . Both are per seed and aggregated by the median over seeds. A cell is counted only if a majority of seeds grok.

B Model selection for the dose-response law

For a dose response of n cells with held norms $\|W\|_i$ and log grokking times $\ell_i = \ln T_i$, each candidate form is fit by least squares and scored by $\text{AIC} = n \ln(\text{RSS}/n) + 2k$ with k the parameter count. The compared forms are the exponential $\ell = c + \alpha \|W\|$ ($k = 2$), a power law with free offset $\ell = c + \beta \ln(\|W\| - w_0)$ ($k = 3$), and a saddle-node form $\ell = c - \frac{1}{2} \ln(w_0 - \|W\|)$, $w_0 > \max \|W\|$ ($k = 2$), the critical-slowing-down signature a fold bifurcation would produce. Only majority-grokked cells on the softmax-collapse-free range ($\rho \leq 1.15$) enter the fit. ΔAIC in Table 1 is the

exponential’s margin over the better of the power-law and saddle-node forms; positive favors the exponential.

C Mediation: per-cell data and the recovery fraction

Tables 5–6 give every cell behind §4.1 (medians over 12 seeds; all cells grok 12/12). Each block holds the baseline ($\rho = 1.0$, $\tau = 1$) and the norm-up sweep ($\rho = 1.15$, τ increasing, which lowers the effective logit scale at fixed norm). The logit-scale recovery fraction is $(T_1 - T_2)/(T_1 - T_0)$, where T_0 is the baseline delay, T_1 the norm-up delay at $\tau = 1$, and T_2 the delay read where the effective logit scale, interpolated over the τ sweep, equals the baseline value. The 95% CI is a paired bootstrap over the twelve seeds (4000 resamples): each resample recomputes the per-cell median T_{grok} and the interpolated T_2 , with the effective-logit curve held at its full-sample value; the interval is the 2.5–97.5 percentile range. It therefore reflects seed variability in T_{grok} , the dominant source, and not interpolation uncertainty on the logit axis.

Table 5: Temperature sweep, cross-entropy. L = effective logit scale at grok. All cells grok 12/12.

p	cell	T_{grok}	T_{mem}	delay	L
59	$\rho=1.0, \tau=1$ (baseline)	3675	175	3500	23.87
59	$\rho=1.15, \tau=1.00$	12600	175	12425	34.31
59	$\rho=1.15, \tau=1.15$	9100	175	8925	29.46
59	$\rho=1.15, \tau=1.30$	6825	175	6650	26.50
59	$\rho=1.15, \tau=1.50$	4700	175	4525	23.13
59	$\rho=1.15, \tau=1.70$	3450	200	3250	20.89
97	$\rho=1.0, \tau=1$ (baseline)	1775	200	1575	31.71
97	$\rho=1.15, \tau=1.00$	6050	200	5850	43.01
97	$\rho=1.15, \tau=1.15$	4200	225	3975	38.80
97	$\rho=1.15, \tau=1.30$	3050	225	2825	35.10
97	$\rho=1.15, \tau=1.50$	2188	225	1962	31.49
97	$\rho=1.15, \tau=1.70$	1762	250	1512	29.10

Table 6: Temperature sweep, mean-squared error. The effective logit scale L is pinned near baseline and does not move with τ , so no recovery fraction is defined. All cells grok 12/12.

p	cell	T_{grok}	T_{mem}	delay	L
59	$\rho=1.0, \tau=1$ (baseline)	1175	250	925	0.95
59	$\rho=1.15, \tau=1.00$	2225	350	1875	1.04
59	$\rho=1.15, \tau=1.15$	2088	338	1750	1.02
59	$\rho=1.15, \tau=1.30$	1900	325	1575	0.99
59	$\rho=1.15, \tau=1.50$	1700	325	1375	0.95
59	$\rho=1.15, \tau=1.70$	1550	325	1225	0.93
97	$\rho=1.0, \tau=1$ (baseline)	850	600	250	0.59
97	$\rho=1.15, \tau=1.00$	1412	975	438	0.65
97	$\rho=1.15, \tau=1.15$	1312	888	425	0.60
97	$\rho=1.15, \tau=1.30$	1225	825	400	0.62
97	$\rho=1.15, \tau=1.50$	1150	775	375	0.65
97	$\rho=1.15, \tau=1.70$	1088	750	338	0.57

Under cross-entropy the recovery fraction is 0.83 at $p = 59$ (CI [0.82, 0.85]) and 0.89 at $p = 97$

(CI [0.88, 0.91]). The memorization control reads directly from these tables: T_{mem} moves by 25–50 steps across the full τ sweep while the delay moves by thousands, so the entire τ effect is on the delay. Under mean-squared error the effective logit scale moves only $\sim 1.1\times$ across τ (against $\sim 1.6\times$ under cross-entropy), so no logit-scale channel is available for τ to act on; at $p = 97$ roughly two thirds of the small τ effect falls on T_{mem} , which is a further reason the mean-squared-error residual is not interpreted.

Collapse grid. Table 7 lists the twelve $\rho \times \tau$ cells per modulus behind the collapse of §4.2 (all grok 12/12). The regression statistics are in Table 3.

Table 7: The $\rho \times \tau$ grid behind Fig. 2: T_{grok} (median over 12 seeds) and the effective logit scale L at grok, cross-entropy. Cells of different ρ but matched L share the delay.

(ρ, τ)	$p = 59$		$p = 97$	
	T_{grok}	L	T_{grok}	L
(1.00, 1.0)	3675	23.9	1775	31.7
(1.00, 1.3)	2125	19.6	1250	26.7
(1.00, 1.7)	1700	16.4	1088	22.9
(1.05, 1.0)	5938	24.9	2650	35.0
(1.05, 1.3)	3050	20.9	1550	29.4
(1.05, 1.7)	1925	17.9	1200	26.7
(1.10, 1.0)	8888	29.3	4038	38.7
(1.10, 1.3)	4600	23.2	2100	31.6
(1.10, 1.7)	2475	19.6	1425	26.8
(1.15, 1.0)	12600	34.3	6050	43.0
(1.15, 1.3)	6825	26.5	3050	35.1
(1.15, 1.7)	3450	20.9	1762	29.1

No-LayerNorm transformer. Table 8 gives the transformer temperature sweep at $p = 59$ (all grok 12/12). At the raised norm, increasing τ shortens the delay monotonically by $4.7\times$. The effective logit scale at grok is large and non-monotonic in τ (the highest-stress $\tau = 1.0$ cell exhibits post-grokking logit growth past 1000), so we report this as qualitative corroboration and do not compute a recovery fraction for the transformer.

Table 8: No-LayerNorm transformer, cross-entropy, $p = 59$ (medians over 12 seeds). The delay responds to the held norm and to τ as in the MLP; the effective logit scale L is larger and noisier.

cell	T_{grok}	L at grok	grok
$\rho=1.0, \tau=1$ (baseline)	1238	208	12/12
$\rho=1.15, \tau=1.00$	16112	165	12/12
$\rho=1.15, \tau=1.15$	9525	153	12/12
$\rho=1.15, \tau=1.30$	6525	138	12/12
$\rho=1.15, \tau=1.50$	4712	164	12/12
$\rho=1.15, \tau=1.70$	3462	177	12/12

D Softmax-collapse audit and the layer-allocation control

Softmax-collapse audit. The audit of §5 re-runs the two highest-dose cross-entropy cells at $p = 59$ in float64, at held norms identical to the float32 run, and records the softmax-collapse rate (the

fraction of training points whose correct-class softmax probability reaches one in working precision). Details in Table 4.

Layer allocation (Arm A). As a second intervention we hold the total norm fixed at $\rho = 1.0$ ($\|W\| = w_c$) and shift mass between the embedding and the readout by a parameter γ : the readout share of the E - W_2 power budget is scaled by $(1 + \gamma)$ (clipped), with W_1 held at its baseline share, so $\gamma > 0$ moves mass into W_2 at fixed total norm. Table 9 reports the result. The grokking time changes by $2.2\times$ ($p = 59$) and $1.7\times$ ($p = 97$) across the sweep at fixed total norm, which corroborates that the scalar norm is not the sole driver. We treat it as supporting, not primary for two reasons: the effective logit scale at grok barely moves across γ (it is the *total* readout magnitude, not its share, that the temperature test varies cleanly), so the change is driven by the embedding-capacity shift rather than by a clean logit-scale move; and at $\gamma = +0.6$, $p = 59$, three of twelve seeds fail to grok within budget. The temperature test of §4.1 is the clean manipulation.

Table 9: Layer allocation at fixed total norm ($\rho = 1.0$). $\gamma > 0$ shifts mass into the readout W_2 . L = effective logit scale at grok. Medians over 12 seeds unless the grok count notes otherwise.

p	γ	T_{grok}	$\ W\ $	$\ E\ $	$\ W_2\ $	L
59	-0.60	2975	54.5	28.4	15.6	20.60
59	-0.30	4012	54.5	25.0	20.6	23.35
59	+0.00	3962	54.5	21.0	24.7	24.46
59	+0.30	2862	54.5	16.1	28.1	24.01
59	+0.60	1825 [†]	54.5	8.8	31.2	24.84
97	-0.60	1462	65.9	39.0	21.1	27.35
97	-0.30	1950	65.9	34.5	27.9	31.34
97	+0.00	1975	65.9	29.3	33.3	32.43
97	+0.30	1538	65.9	22.9	38.0	31.91
97	+0.60	1150	65.9	13.8	42.1	32.62

[†] 9 of 12 seeds grok within budget; the median is over the grokked seeds.

E Same-state arms

Table 10 gives the four arms forked from one identical state at $t = 800$ (§4.6), all grokking 12/12. The delay is measured from the fork. The recovery contrast is clamp-at- N_0 against clamp-at- ρN_0 , both applying the same clamp operation from the same state.

Table 10: Same-state arms, cross-entropy, fork at $t = 800$ (median delay from fork over 12 seeds). The raised/same ratio is a paired bootstrap over the 12 seeds. The monotonic ordering holds in all 12 seeds at both moduli.

arm (held norm)	delay $p = 59$	delay $p = 97$
clamp-lowered (N_0/ρ)	1525	1338
free (norm decays)	4225	1950
clamp-same (N_0)	7200	6600
clamp-raised (ρN_0)	24162	21600
raised/same ratio	3.36 [3.27, 3.43]	3.27 [3.21, 3.38]

Code and data availability

The clamp and temperature-mediation runner, the per-cell metric outputs used for every fit, table, and figure, the AIC model-selection script, and the float64 softmax-collapse audit are available at <https://github.com/ClevixLab/grokking-logit-scale>, with seeds fixed so that every number in Tables 1–4 and the figures can be regenerated.

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